

The ubiquitin-conjugating enzyme UBE2D/eff maintains a youthful proteome and ensures protein quality control during aging

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Abstract

Summary

Ubiquitin-conjugating enzymes (E2s) are key for regulating protein function and turnover via ubiquitination but it remains undetermined which E2s maintain proteostasis during aging. Here, we find that E2s have diverse roles in handling a model aggregation-prone protein (huntingtin-polyQ) in the *Drosophila* retina: while some E2s mediate aggregate assembly, UBE2D/effete (eff) and other E2s are required for huntingtin-polyQ degradation. UBE2D/eff is key for proteostasis also in skeletal muscle: eff protein levels decline with aging, and muscle-specific eff knockdown causes an accelerated buildup in insoluble poly-ubiquitinated proteins (which progressively accumulate with aging) and shortens lifespan. Transgenic expression of human UBE2D2, homologous to eff, partially rescues the lifespan and proteostasis deficits caused by muscle-specific eff^{RNAi} by re-establishing the physiological levels of eff^{RNAi}-regulated proteins, which include several regulators of proteostasis. Interestingly, UBE2D/eff knockdown in young age reproduces part of the proteomic changes that normally occur in old muscles, suggesting that the decrease in UBE2D/eff protein levels that occurs with aging contributes to reshaping the composition of the muscle proteome. Altogether, these findings indicate that UBE2D/eff is a key E2 ubiquitin-conjugating enzyme that ensures protein quality control and helps maintain a youthful proteome composition during aging.

eLife assessment

This **valuable** study presents findings on the role of the ubiquitin-conjugating enzyme UBE2D/eff in maintaining proteostasis during aging. The evidence supporting the conclusions is **solid**, although one reviewer had concerns about the readout for protein aggregation and the loss-of-function studies. In the future, mechanistic insights explaining the impact of UBE2D/eff deficiency on the accumulation of poly-ubiquitinated proteins and in shortening lifespan would be interesting. The present study is of broad interest to cell biologists working in aging and age-related diseases.

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Introduction

Protein degradation regulates many cellular functions, and its derangement is the underlying cause of many human diseases (1–4). Most proteins are regulated via degradation to ensure dynamic adjustments in their concentrations in response to cellular challenges and to maintain physiologic homeostasis. Moreover, damaged and misfolded proteins are degraded to avoid toxicity caused by their interaction with native proteins (1–4). Because of these important functions, protein degradation is tightly controlled. In most cell types, the ubiquitin/proteasome system (UPS) is responsible for most of the protein degradation in the nucleus and cytoplasm whereas the autophagy/lysosome system degrades cellular organelles, protein aggregates, and long-lived proteins (1–6). In addition, several proteases and peptidases cooperate with the proteasome and autophagy in degrading target proteins (7–10).

To route proteins for degradation, the UPS relies on ubiquitination, a post-translational modification that also regulates protein function and localization (11, 12). Through the concerted actions of a single ubiquitin-activating enzyme (E1), ~35 ubiquitin-conjugating enzymes (E2s), and ~620 E3 ubiquitin ligases, the ubiquitin-proteasome system (UPS) orchestrates the specific poly-ubiquitination of protein substrates in humans, which can lead to their degradation (6, 13–16). Depending on the ubiquitin lysine residue employed to build poly-ubiquitin chains, poly-ubiquitinated proteins can be preferentially degraded by the proteasome or by the autophagy-lysosome system (11, 12, 17–21). E2s have a pivotal role in the ubiquitination cascade and direct the recruitment of E3 ubiquitin ligases and target proteins (21–24).

In addition to regulating normal protein turnover, E2s can also ensure proteostasis. For example, the E2 enzyme UBE2B and its associated E3 ligase UBR4 have been found to regulate muscle protein quality during aging in *Drosophila* and mice (25, 26). The capacity of E2s to maintain proteostasis may derive from their capacity to guide the ubiquitination and proteasome-mediated degradation of misfolded and aggregation-prone proteins (27, 28) and to trigger autophagy (29, 30). Moreover, because some E2s are physically associated with the proteasome, it has been proposed that they may directly regulate its proteolytic activity (31–33). For example, the E2 enzymes Ubc1/2/4/5 interact with the proteasome and this association further increases with heat stress in yeast (31), a condition where proteasome function is modulated by ubiquitination of proteasome components, such as Rpn13, mediated by the proteasome-associated E3 UBE3C (34). While E2s may regulate proteostasis via several mechanisms, it remains largely undetermined which E2s are crucial for maintaining protein quality control during aging.

Here, we have utilized a model aggregation-prone protein (GFP-tagged pathogenic huntingtin, Htt-polyQ-GFP) expressed in the *Drosophila* retina to test the impact of E2 ubiquitin-conjugating enzymes on proteostasis. In addition, we have similarly tested a set of E3 ubiquitin ligases and deubiquitinating enzymes (DUBs) that were identified by mapping the E2 interactome in human cells (35). These analyses reveal diverse functions of E2s and associated enzymes in proteostasis. Contrary to the expectation that E2s may generally promote the proteasomal degradation of Htt-polyQ and hence reduce Htt-polyQ levels, we find that the knockdown of several E2s has the opposite effect, i.e. it decreases Htt-polyQ aggregates.

Knockdown of the E2 enzyme effete (eff), homologous to human UBE2D1/2/3/4 ubiquitin-conjugating enzymes, is one of the relatively few RNAi interventions that increase the amount of Htt-polyQ aggregates, and we find that this is due to an impediment in Htt-polyQ protein degradation. Consistent with a prevalent role of UBE2D/eff in proteostasis across aging tissues in *Drosophila*, we find that UBE2D/eff protein levels significantly decline during aging in skeletal muscle, and that muscle-specific eff knockdown increases the age-associated accumulation of insoluble poly-ubiquitinated proteins and reduces lifespan. Interestingly, the proteostasis deficits caused by UBE2D/eff knockdown are partially rescued by transgenic expression of one of its human homologs, UBE2D2, suggesting that UBE2D ubiquitin-conjugating enzymes have evolutionary conserved roles in maintaining protein quality control during aging. Altogether, our study identifies diverse roles for E2 ubiquitin-conjugating enzymes in handling aggregation-prone proteins and managing proteostasis, and that UBE2D/eff is key for ensuring protein quality control during aging.

Results

Diverse roles of E2 ubiquitin-conjugating enzymes in the disposal of pathogenic huntingtin

Huntington disease (HD) is caused by pathogenic huntingtin with polyQ tract expansion (Htt-polyQ), a toxic, aggregation-prone protein that induces neurodegeneration (36–38). Previous studies have shown that ubiquitination is a conserved regulator of polyglutamine aggregation and that E2 loss can both increase and decrease the size and number of Htt-polyQ aggregates (25, 26, 39, 40). However, as for other analyses of E2 function, a complete understanding of the role of E2s in polyglutamine aggregation is missing.

The fruit fly *Drosophila melanogaster* is an ideal model for studying the mechanisms of Htt-polyQ protein degradation. GFP-tagged Htt-polyQ72 can be expressed in the fly retina (41) and fluorescent Htt-polyQ-GFP aggregates can be visualized from intact animals over time (42, 43). In particular, the fly retina is a convenient tissue to examine Htt-polyQ because this tissue is dispensable for fly survival. With this model, it was previously found that there is an age-dependent increase in the amount of Htt-polyQ72 protein aggregates, which reflects the overall age dependency of HD phenotypes also observed in humans (36, 37).

To test whether E2s regulate Htt-polyQ-GFP aggregation during aging, multiple RNAi lines were utilized to target each of the 21 *Drosophila* E2s. For these studies, transgenic RNAi was driven in the *Drosophila* retina with the UAS/Gal4 system and GMR-Gal4, concomitantly with Htt-polyQ-GFP (Fig. 1A). After 30 days at 29°C, the total area of Htt-Q72-GFP fluorescent aggregates was scored with CellProfiler and compared to control RNAi interventions (Fig. 1A–C and Supplementary Table 1). Overall, there was a trend towards decreased aggregate area upon knockdown of several E2s, including the *Drosophila* homologs of UBE2F/M, UBE2I, UBE2QL1, and UBE2Z, suggesting that these E2s may be required for ubiquitination-dependent aggregation of Htt-polyQ-GFP (Fig. 1A–B). Analyses of deubiquitinating enzymes (DUBs) and E3 ubiquitin ligases identified from mapping the E2 interactome in human cells (35) indicate that RNAi for these interactors yields

overall similar phenotypes as those found by knocking down the interacting E2s (**Fig. 1A-C**). For example, RNAi for E3s and DUBs that interact with Ubc10 (UBE2L3) and CG7656 (UBE2R1/2) reduce the area of Htt-polyQ-GFP aggregates, as found for knockdown of these E2s (**Fig. 1C**). The decreased polyglutamine aggregation that is observed upon E2 RNAi may indicate that E2-mediated ubiquitination is required for Htt aggregate formation. Consistent with this model, it was previously reported that K48 linkage-specific ubiquitination drives Htt degradation by the proteasome whereas K63 ubiquitination promotes Htt aggregation (44). Our comprehensive analysis now highlights the differential effects of distinct E2s on Htt disposal and suggests that this may occur because of the specificity in the E3s and DUBs interacting with each E2.

UBE2D/eff knockdown increases aggregates of pathogenic huntingtin in the *Drosophila* retina

In addition to RNAi lines that decrease the amount of Htt aggregates, we found that RNAi-mediated knockdown of other E2s causes an increase in Htt aggregates (**Fig. 1A-C**). Specifically, RNAi for Ubc6 (UBE2A/B), eff (UBE2D), Ubc10 (UBE2L3), and Ubc84D (UBE2L6) increased the total area of protein aggregates. Likewise, RNAi for E3/DUBs that physically associate with these E2s (such as Ubr4 and Kcfm1, associated with UBE2A/B; and HUWE1, associated with UBE2D) led to higher levels of polyglutamine aggregates. A possible interpretation of these results is that these E2s are necessary for the degradation of aggregation-prone Htt and, consequently, their loss leads to the accumulation of detergent-insoluble polyglutamine aggregates. To test this hypothesis, we further characterized the molecular mechanisms by which eff/UBE2D regulates Htt proteostasis. We first measured by qRT-PCR the levels of Htt-polyQ-GFP and found that there are no changes in its expression upon eff RNAi compared to controls (**Fig. 1D**), indicating that increased levels of Htt-polyQ-GFP aggregates observed upon eff RNAi do not arise from increased transgenic expression of Htt-polyQ-GFP. Next, we utilized western blot to analyze the levels of Htt-polyQ-GFP monomers (~50 kDa) and high-molecular-weight (HMW) Htt-polyQ-GFP aggregates that accumulate in the stacking gel (>250 kDa). Remarkably, 2 different RNAi lines targeting eff/UBE2D increase HMW Htt-polyQ-GFP levels (**Fig. 1E**), suggesting that eff/UBE2D-mediated ubiquitination may be required for Htt degradation. In summary, these findings indicate an extensive role of E2 ubiquitin-conjugating enzymes in modulating Htt-Q72-GFP aggregation during aging in the *Drosophila* retina.

Retinal degeneration caused by UBE2D/eff knockdown is rescued by human UBE2D2/4

Compared to controls, we find that RNAi for Ubc6 and for eff induces retinal degeneration, as indicated by the “rough eye” appearance (indicative of photoreceptor death and derangement of inter-ommatidial bristles (45, 46)), the increase in the eye area occupied by necrotic patches, and/or the loss of pigmentation (**Fig. 2A**). On this basis, we next examined whether concomitant transgenic expression of the human homologs of these *Drosophila* E2s rescues retinal degeneration. On this basis, we expressed human UBE2B (homologous to Ubc6, DIOPT homology score=13) and human UBE2D2 and UBE2D4 (homologous to eff, DIOPT scores of 13 and 10, respectively) and compared their capacity to rescue retinal degeneration compared to mock mcherry overexpression (**Fig. 2A**). As expected based on the sequence homology, hUBE2B (but not mcherry, hUBE2D2, and hUBE2D4) rescued retinal degeneration (i.e. necrotic patches and depigmentation) induced by Ubc6 RNAi. Similarly, hUBE2D2 and to a lower extent hUBE2D4 rescued depigmentation caused by eff knockdown. Altogether, these findings indicate that Ubc6 and eff loss can be rescued by transgenic expression of their respective human homologs UBE2B and UBE2D2/4 (**Fig. 2A**).

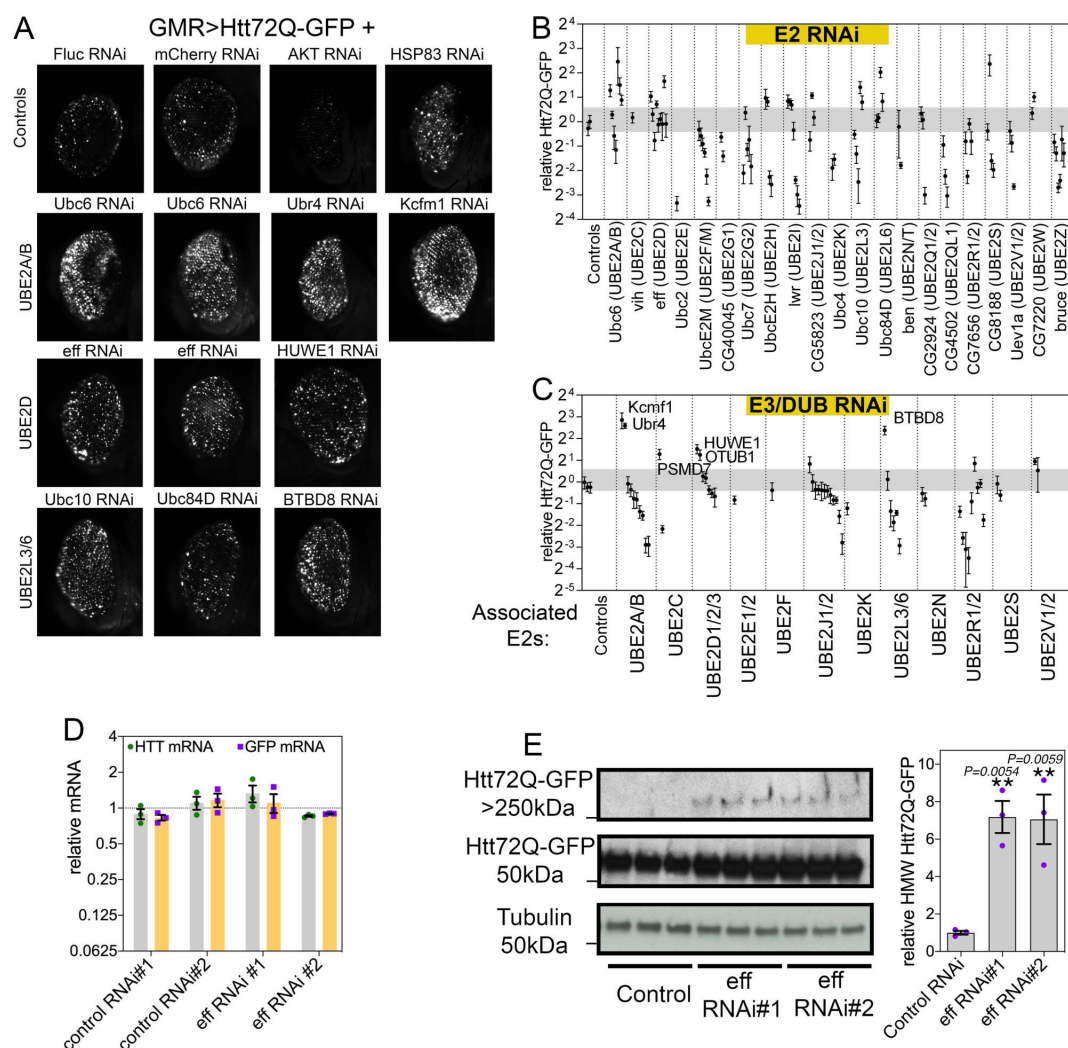


Fig. 1.

RNAi for E2 ubiquitin-conjugating enzymes and associated E3s and DUBs modulates pathogenic huntingtin-polyQ aggregates in the *Drosophila* retina

(A) GFP-tagged huntingtin-polyQ (Htt-polyQ72-GFP) driven with *GMR-Gal4* leads to GFP-fluorescent Htt protein aggregates in the retina at 30 days of age. Compared to negative controls (mCherry^{RNAi} and luciferase^{RNAi}), RNAi for E2 ubiquitin-conjugating enzymes, associated deubiquitinating enzymes (DUBs), and E3 ubiquitin ligases (E3s) modulates the amount of Htt-polyQ72-GFP aggregates. Specifically, RNAi for Ubc6 (homologous of UBE2A/B) and for its associated E3 ubiquitin ligases Ubr4 and Kcmf1 increases the amount of Htt-polyQ72-GFP protein aggregates. Similar increases are also seen with knockdown of eff/UBE2D and the associated E3 HUWE1, and with RNAi for Ubc10/UBE2L3, Ubc84D/UBE2L6, and for the associated E3 enzyme BTBD8. Positive controls include Akt RNAi, which reduces protein aggregates, and Hsp83 RNAi, which increases them. (B-C) Quantitation of the total area of Htt-polyQ72-GFP aggregates modulated by RNAi for E2s (B) and for associated E3s and DUBs (C). Relative fold changes compared to control RNAi interventions are shown; n=5 (biological replicates), SD. Each data point in the graph represents a single RNAi targeting the corresponding gene. For each RNAi, the mean of 5 biological replicates is shown, with each biological replicate representing a single eye (each from a distinct animal). (D) qRT-PCR indicates that there are no changes in *GFP* and *Htt* mRNA levels upon eff knockdown compared to control RNAi, indicating that eff RNAi does not modulate the amount of GFP-tagged huntingtin-polyQ aggregates via changes in the expression of *Htt-polyQ72-GFP* transgenes; n=3 (biological replicates) and SEM. (E) Levels of Htt-polyQ72-GFP aggregates detected by western blot with anti-GFP antibodies identify Htt-polyQ72-GFP monomers (~50 kDa) and high-molecular-weight (HMW) assemblies of Htt-polyQ72-GFP in the stacking gel (>250 kDa). Knockdown of eff/UBE2D increases the levels of HMW Htt-polyQ72-GFP; n=3 (biological replicates), SEM, and *p*-values (one-way ANOVA) are indicated, with ***p*<0.01, compared to mcherry^{RNAi}.

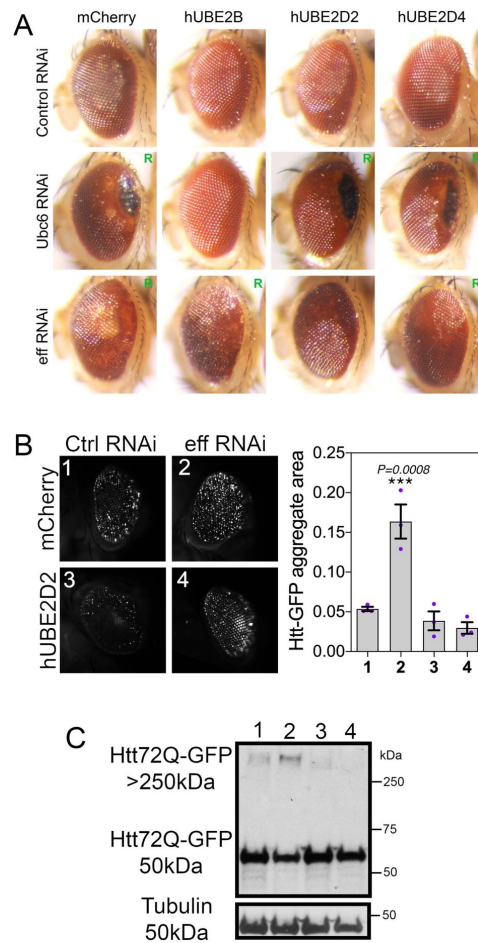


Fig. 2.

Retinal degeneration induced by knockdown of Ubc6 and eff is rescued by expression of the respective human homologs UBE2B and UBE2D2/4

(A) Compared to control RNAi, knockdown of Ubc6 and eff causes retinal degeneration, as indicated by the “rough eye” phenotype (R), by black necrotic patches, and/or by areas with depigmentation. Human UBE2B rescues degeneration induced by RNAi for its *Drosophila* homolog Ubc6 but not induced by RNAi for the unrelated E2 enzyme eff. Likewise, human UBE2D2 and UBE2D4 rescue retinal degeneration induced by knockdown of their *Drosophila* homolog eff. (B) Htt-polyQ72-GFP aggregates increase in *Drosophila* retinas with eff RNAi and this is rescued by human UBE2D2 but not by mcherry; n=3 (biological replicates), SEM, and *p*-values (one-way ANOVA) are indicated, with ****p*<0.001. (C) Similarly, high-molecular-weight assemblies of Htt-polyQ72-GFP detected in the stacking gel increase with eff RNAi compared to control RNAi but this is rescued by hUBE2D2 versus control mcherry expression.

On this basis, we next examined whether hUBE2D can rescue the defects in Htt-polyQ proteostasis caused by eff RNAi. Specifically, we compared eff RNAi versus control RNAi, with the concomitant expression of either mcherry or hUBE2D2. Image analysis of Htt-poly-GFP aggregates indicates that hUBE2D2 impedes the accumulation of Htt-polyQ-GFP aggregates which are otherwise increased by eff RNAi (**Fig. 2B**). Similar results were also obtained by western blot: there was lower accumulation of HMW Htt-polyQ-GFP in eff^{RNAi}+hUBE2D2 versus eff^{RNAi}+mcherry and the other controls (**Fig. 2C**). In summary, these findings indicate that eff/UBE2D has an evolutionary-conserved function in maintaining proteostasis in a Huntington's disease model.

UBE2D/eff protein levels decline with aging, and UBE2D/eff knockdown impairs skeletal muscle proteostasis as observed during aging

The comprehensive analysis reported above has revealed a range of diverse roles for E2s in modulating polyglutamine aggregates in *Drosophila*. UBE2D/eff appears to be one of the E2s with the most striking effects in preserving proteostasis, i.e. it is required for preventing the accumulation of HMW Htt-polyQ-GP (**Fig. 1**–**2**). On this basis, we further probed the role of UBE2D/eff in protein quality control. To this purpose, we examined skeletal muscle aging, which is characterized by a progressive decline in proteostasis and by the age-related accumulation of detergent-insoluble poly-ubiquitin protein aggregates (25, 47–53).

Deep-coverage tandem mass tag (TMT) mass spectrometry (54, 55) of skeletal muscle from young and old control w¹¹¹⁸ flies (1 and 8 weeks old) detected 5,971 proteins, including 10 of the 21 *Drosophila* E2 ubiquitin-conjugating enzymes (**Fig. 3A** and Supplementary Table 2). Interestingly, UBE2D/eff displayed the strongest age-dependent downregulation (**Fig. 3B**), suggesting that a reduction in UBE2D/eff protein levels may contribute to the progressive loss of protein quality control with aging.

To test this hypothesis, we reduced UBE2D/eff levels from young age by driving UBE2D/eff RNAi with MhcF3-Gal4 (25) in skeletal muscle, and compared this intervention to a control mcherry RNAi. Immunostaining and confocal microscopy of skeletal muscle of flies at 3 weeks of age indicate that UBE2D/eff RNAi significantly increases the size of poly-ubiquitin protein aggregates and that there is an overall trend towards a higher total area of aggregates, compared to control RNAi (**Fig. 3C–D**). Altogether, these results indicate that anticipating the age-related decline in eff/UBE2D protein levels deranges proteostasis.

We next examined muscle proteostasis via the analysis of detergent-soluble and insoluble fractions (56). Knockdown of UBE2D/eff with 2 distinct RNAi lines increased the levels of poly-ubiquitinated proteins found in both the soluble and insoluble fractions of skeletal muscle of 3-weeks-old flies. There was also an increase in Ref(2)P/p62 levels that occurred in parallel with the accumulation of poly-ubiquitinated proteins, although this occurred primarily in the detergent-soluble fraction (**Fig. 3E–F**). Altogether, these findings indicate that UBE2D/eff is required to ensure protein quality control in skeletal muscle.

Defects in proteostasis and lifespan caused by muscle-specific UBE2D/eff knockdown are partially rescued by transgenic expression of human UBE2D2

We have found that human UBE2D2 can rescue retinal degeneration induced by eff RNAi in the context of polyglutamine disease, and that this results from the degradation of HMW Htt-polyQ (**Fig. 2**). On this basis, we next examined whether hUBE2D2 can also rescue defects in muscle proteostasis due to eff knockdown. To this purpose we examined detergent-soluble and insoluble fractions from muscles of flies with eff^{RNAi}+hUBE2D2 versus eff^{RNAi}+mcherry as well as

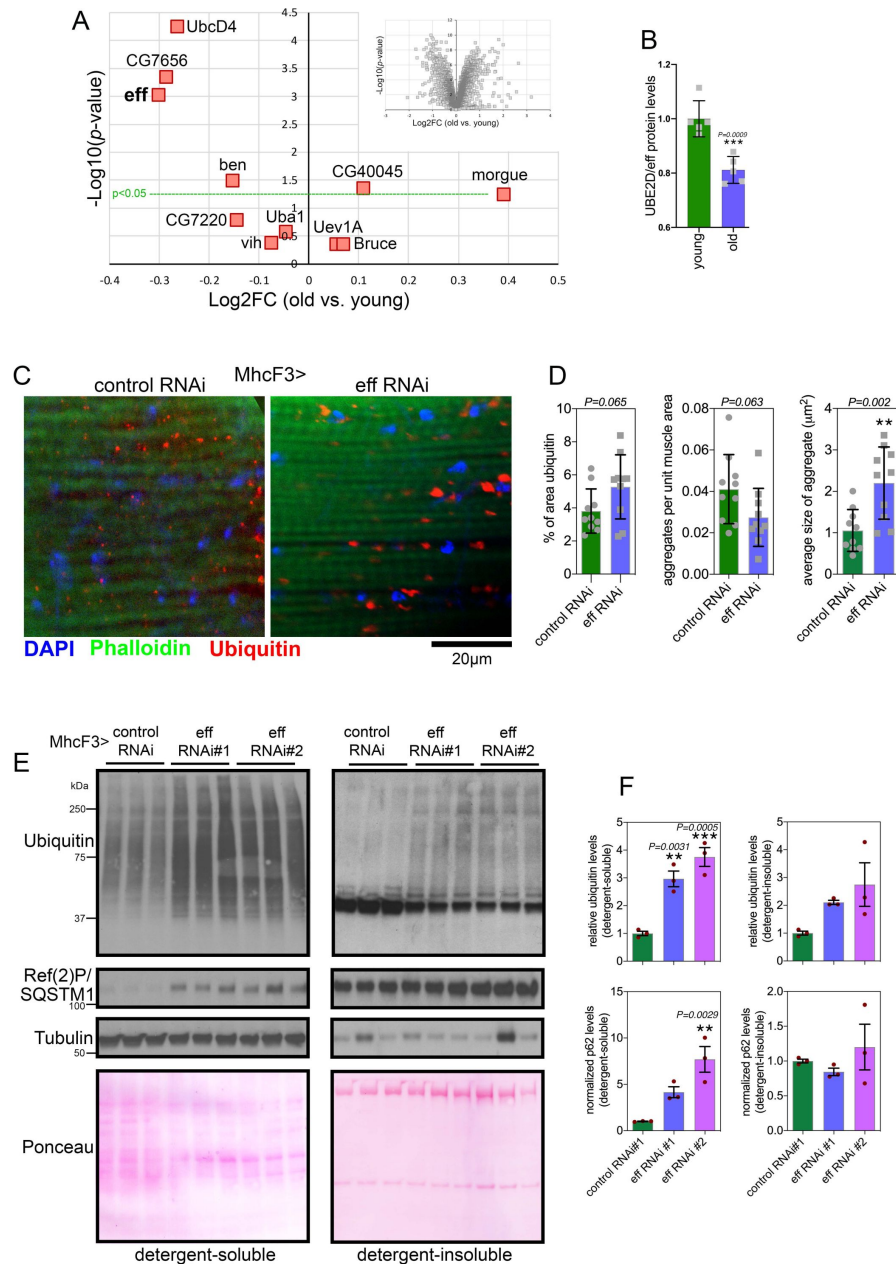


Fig. 3.

RNAi for the E2 enzyme eff/UBE2D impairs muscle protein quality control during aging

(A) TMT mass spectrometry of *Drosophila* skeletal muscle from old versus young w^{1118} flies at 8 vs. 1 week old; $n=5$ biological replicates. The x axis reports the Log2FC whereas the y axis reports the $-\log_{10}(p\text{-value})$. The E1 enzyme UBA1 and the E2 enzymes that are detected are indicated, including UBE2D/eff; $n=5$ (biological replicates).

(B) The protein levels of UBE2D/eff significantly decline with aging (8 versus 1 week-old) in skeletal muscle; $n=5$ (biological replicates), SD, and p -values (Student's t -test).

(C-D) Immunostaining of *Drosophila* skeletal muscle at 3 weeks of age indicates that eff/UBE2D RNAi impairs protein quality control, as indicated by the higher age-related accumulation of aggregates of poly-ubiquitinated proteins, compared to control RNAi. The scale bar is 20 μm . In (C), $n=10$ (biological replicates), SD, and p -values (Student's t -test).

(E-F) Western blot analysis of detergent-soluble and insoluble fractions from *Drosophila* skeletal muscle indicates that eff RNAi impedes proteostasis, as indicated by higher levels of detergent-soluble and insoluble poly-ubiquitinated proteins compared to control RNAi (E). A similar increase is also found for the detergent-soluble levels of Ref(2)P/p62 (E). $n=3$ (biological replicates), SEM, $***p < 0.01$, ns=not significant (one-way ANOVA).

control^{RNAi}+hUBE2D2 versus control^{RNAi}+mcherry (**Fig. 4A-B** [↗](#)). Western blot analyses with anti-ubiquitin and anti-p62/Ref(2)P antibodies revealed that eff^{RNAi}+hUBE2D2 reduced the detergent-soluble levels of ubiquitinated proteins and p62/Ref(2)P compared to eff^{RNAi}+mcherry and other controls, whereas there was no rescue of detergent-insoluble levels (**Fig. 4A-B** [↗](#)).

Because muscle proteostasis has been found to regulate organismal aging ([25](#) [↗](#), [47](#) [↗](#), [49](#) [↗](#), [57](#) [↗](#), [58](#) [↗](#)), we next determined the impact of muscle-specific UBE2D/eff loss on lifespan. Knockdown of eff with 2 distinct RNAi lines (**Fig. 4C** [↗](#)) shortened lifespan (**Fig. 4D** [↗](#)) and this was partially rescued by concomitant expression of hUBE2D but not by control mcherry (**Fig. 4E** [↗](#)). Altogether, these studies identify a key role for UBE2D/eff in proteostasis during skeletal muscle aging in *Drosophila*, and highlight the evolutionary conservation of UBE2D function.

Deep-coverage proteomics reveals protein substrates that are modulated by eff/UBE2D

To determine the proteins that are modulated by UBE2D/eff RNAi and the partial rescue of proteostasis defects by hUBE2D, TMT-based proteomics ([54](#) [↗](#), [55](#) [↗](#)) was utilized to determine the protein changes modulated by eff/UBE2D RNAi versus control RNAi, and whether these are rescued by hUBE2D2 compared to control mcherry expression ([35](#) [↗](#)). Analysis of eff-induced protein changes revealed several categories that are enriched among the significantly regulated proteins (**Fig. 5A-D** [↗](#) and Supplementary Table 3; ([35](#) [↗](#))). Upregulated proteins included proteasome components, chaperones, deubiquitinases, ubiquitin ligases, and extracellular proteins whereas downregulated proteins were enriched for peptidases, lipases, and secreted proteins (**Fig. 5E** [↗](#)).

Curated analyses revealed that, as expected, eff protein levels declined in response to eff RNAi compared to control RNAi (**Fig. 5F** [↗](#)). The levels of several other proteins were increased upon eff RNAi but reduced by concomitant rescue with hUBE2D2 (**Fig. 5F** [↗](#)). These proteins include Arc1 and Arc2 (activity-regulated cytoskeleton-associated protein 1 and 2), which regulate starvation-induced locomotion, the neuromuscular junction, and metabolism ([59](#) [↗](#)–[61](#) [↗](#)); the glycine N-methyltransferase Gnmt enzyme, which generates sarcosine and controls the amount of the methyl donor S-adenosylmethionine ([62](#) [↗](#), [63](#) [↗](#)); and CG4594, which encodes for an enzyme involved in fatty acid beta oxidation. In addition to rescuing protein changes induced by eff RNAi, overexpression of hUBE2D2 by itself reduced the levels of Arc1, Arc2, Gnmt, and CG4594. Altogether, these proteomics analyses identify protein targets that are modulated by eff/UBE2D in skeletal muscle (**Fig. 5F** [↗](#)). Improper eff/UBE2D-mediated turnover of these proteins and their consequent accumulation may contribute to derange muscle protein quality control and in turn to reduced survival. In agreement with this hypothesis, it was previously found that elevation of Arc1 protein levels in Alzheimer's disease models causes cytotoxicity and contributes to neuronal death in *Drosophila* ([64](#) [↗](#)).

Conversely, there were other proteins that were induced in muscles with eff/UBE2D knockdown and that may represent a protective response that mitigates the derangement of proteostasis due to eff/UBE2D knockdown. These included Pomp, involved in proteasome assembly ([65](#) [↗](#), [66](#) [↗](#)), the chaperones Hsp22, Hsp23, and Hsp26, and the ubiquitin-like protein UBL3 (**Fig. 5G** [↗](#)).

In summary, these studies indicate that eff/UBE2D knockdown induces several proteomics changes, some of which are rescued by hUBE2D2. While some of these changes may be cytotoxic (e.g. increased Arc1 levels, ([64](#) [↗](#))), others may conversely limit the derangement of protein quality control and/or improve survival.

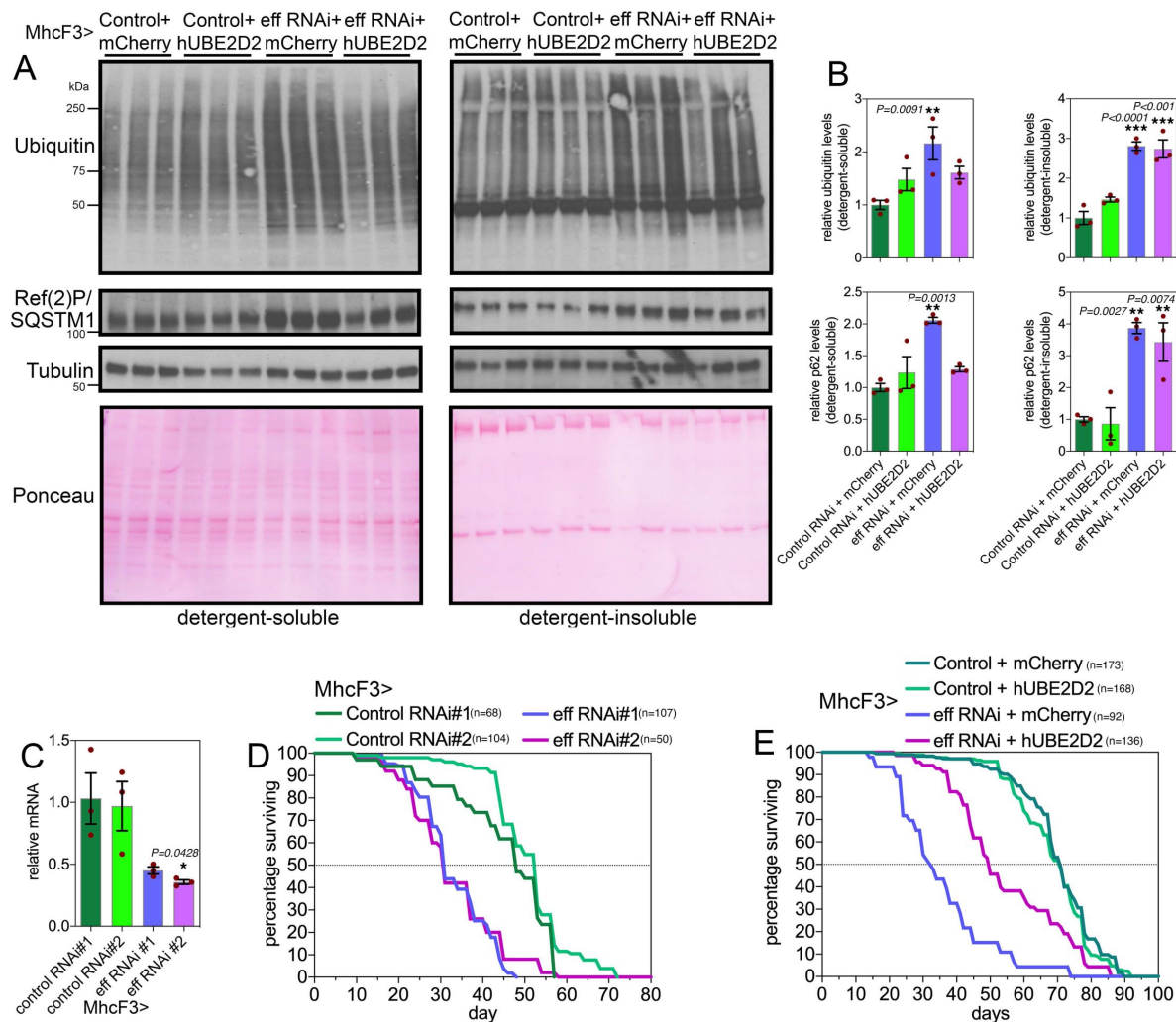


Fig. 4.

Defects in proteostasis and lifespan caused by muscle-specific eff/UBE2D RNAi are partially rescued by the expression of human UBE2D2

(A-B) Western blot analysis of detergent-soluble and insoluble fractions from *Drosophila* skeletal muscle indicates that defects in proteostasis due to eff RNAi can be partially rescued by its human homolog UBE2D2, as indicated by the normalization of the detergent-soluble levels of poly-ubiquitinated proteins and Ref(2)P/p62. However, hUBE2D2 does not impact their detergent-insoluble levels, indicating that hUBE2D2 only partially rescues defects induced by eff/UBE2D RNAi. In (B), $n=3$ (biological replicates), SEM, $**p<0.01$ (one-way ANOVA). (C) Muscle-specific eff/UBE2D RNAi reduces eff mRNA levels; $n=3$ (biological replicates), SEM, $**p<0.01$ (one-way ANOVA). (D) Two distinct RNAi lines targeting eff/UBE2D reduce lifespan compared to control RNAi ($p<0.001$, log-rank test). (E) The decline in organismal survival due to eff/UBE2D RNAi in muscle is partially rescued by transgenic hUBE2D2 expression compared to control mcherry overexpression ($p<0.001$, log-rank test).

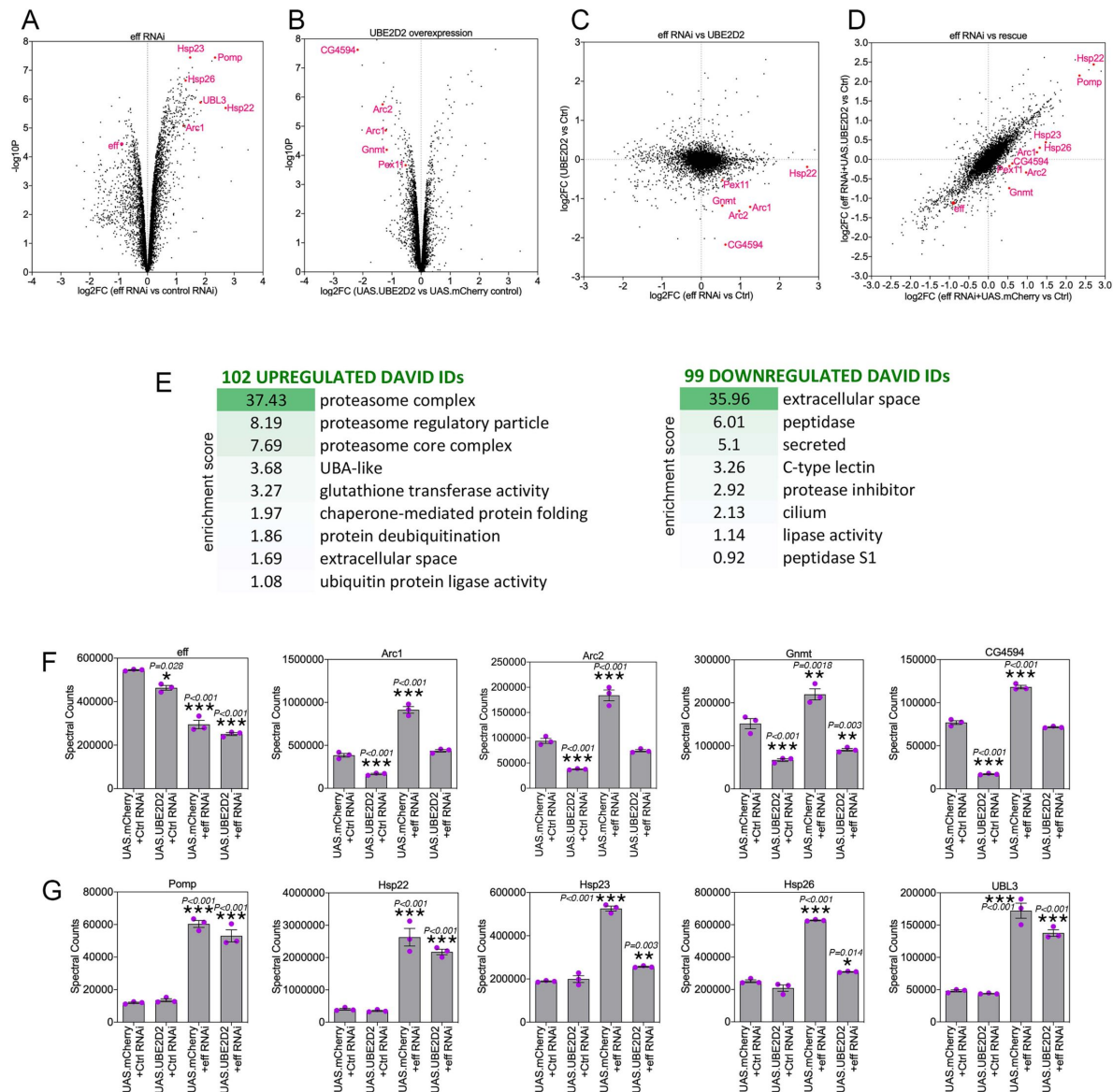


Fig. 5.

Deep-coverage TMT mass spectrometry identifies the proteins modulated by UBE2D/eff knockdown in *Drosophila* skeletal muscle

(A–B) TMT mass spectrometry of *Drosophila* skeletal muscle with eff/UBE2D knockdown versus control RNAi (A) and overexpression of human UBE2D2 versus control mcherry (B). The x axis reports the Log2FC whereas the y axis reports the $-\log_{10}(p\text{-value})$. Examples of regulated proteins are shown in red. (C) Cross-comparison of the Log2FC induced by eff RNAi versus control RNAi (x axis) with the Log2FC of hUBE2D2 versus control mcherry (y axis). (D) Cross-comparison of the Log2FC induced by eff RNAi + mcherry versus control (x axis) with the Log2FC of eff RNAi + hUBE2D2 versus control (y axis) indicates that some of the protein changes induced by eff RNAi are rescued by hUBE2D2. (E) GO term analysis of protein categories that are enriched among upregulated and downregulated proteins in response to eff/UBE2D2 RNAi. The enrichment scores and number of regulated DAVID IDs are shown. (F–G) Examples of regulated proteins modulated by eff RNAi and hUBE2D-mediated rescue include protein changes that may drive derangement of proteostasis, such as Arc1/2 upregulation (F) as well as protein changes that are protective and likely compensatory, such as proteasome components and chaperones (G); $n=3$ (biological replicates), SEM, * $p<0.05$, ** $p<0.01$, *** $p<0.001$ (one-way ANOVA).

RNAi for UBE2D/eff causes proteomic changes associated with aging in *Drosophila* skeletal muscle

We have found that UBE2D/eff protein levels decline during aging in skeletal muscle (Fig. 3A-B) and that the experimental induction of UBE2D/eff knockdown in young age induces a premature decline in proteostasis (Fig. 3C-D), which otherwise is normally lost only in old age (47, 48). Lastly, we have found that RNAi for UBE2D/eff induces many proteomic changes (Fig. 5), consistent with its role in ubiquitin-dependent protein regulation. On this basis, we next examined how the proteomic changes induced by aging correlate with those induced by eff RNAi in young age. To this purpose, we cross-compared the TMT mass spectrometry datasets obtained from *Drosophila* skeletal muscle with eff RNAi (normalized by control mcherry RNAi) and those obtained from control muscles of old versus young flies. Cross-comparison of the significantly regulated genes revealed that 700 out of 1,002 proteins (~70%) are consistently regulated by eff RNAi and aging, whereas the remaining 302 proteins are discordantly regulated (Fig. 6A-D and Supplementary Table 4).

Proteins that were upregulated by both eff RNAi and aging (the largest class, 514 proteins) were enriched for regulators of glutathione metabolism, chaperones, proteasome components, and vesicle transport (Fig. 6B) and included Arc1 and Arc2. Commonly downregulated proteins (186 proteins) included components of mitochondrial ribosomes and late endosomes, SNARE proteins, and proteases/peptidases (Fig. 6C). The discordantly regulated proteins were enriched for oxidoreductases, mitochondrial and endoplasmic reticulum components, and secreted proteins (106 proteins that are upregulated by aging but downregulated by eff RNAi; Fig. 6A), and regulators of cytoplasmic translation and mRNA splicing (196 proteins that are downregulated by aging but upregulated by eff RNAi; Fig. 6D).

Similar results were also obtained with Venn diagrams (Fig. 6E), which identified the proteins with overlapping, consistent, or discordant regulation by aging versus eff^{RNAi}. The threshold of log₂FC >0.3 and <-0.3 was used as a criterion for the selection of significantly up- and down-regulated proteins for these analyses (Fig. 6E). In this case, 23.9% of significantly regulated proteins (141) were upregulated by both aging and eff^{RNAi}, 7.3% (43) were consistently downregulated by both, and 3.2% (19) were oppositely regulated (Fig. 6E).

Altogether, these analyses indicate that UBE2D/eff^{RNAi} causes proteomic changes that recapitulate in part those induced by aging, indicating that UBE2D/eff is necessary to maintain a youthful proteome composition.

Discussion

In this study, we have examined the role of the 21 *Drosophila* E2 ubiquitin-conjugating enzymes in modulating polyglutamine protein aggregation. By analyzing pathogenic huntingtin in the *Drosophila* retina, we found that E2s have diverse functions in regulating Htt-polyQ-GFP. Specifically, the knockdown of many E2s reduces polyglutamine protein aggregates (Fig. 1), suggesting that ubiquitination by these E2s promotes sequestration of pathogenic huntingtin into aggresomes, as found in *C. elegans* (39). Aggregation of Htt has been previously found to decrease its toxicity (67, 68), and therefore E2s that drive Htt aggregation may protect from polyglutamine disease via this mechanism. However, there were also RNAi interventions for other E2s, such as eff/UBE2D, and associated enzymes (DUBs and E3s) that increased the levels of high-molecular-weight assemblies of Htt-polyQ-GFP (Fig. 1C-D), suggesting that these E2s and interacting partners are necessary for Htt-polyQ-GFP degradation. Likewise, eff/UBE2D

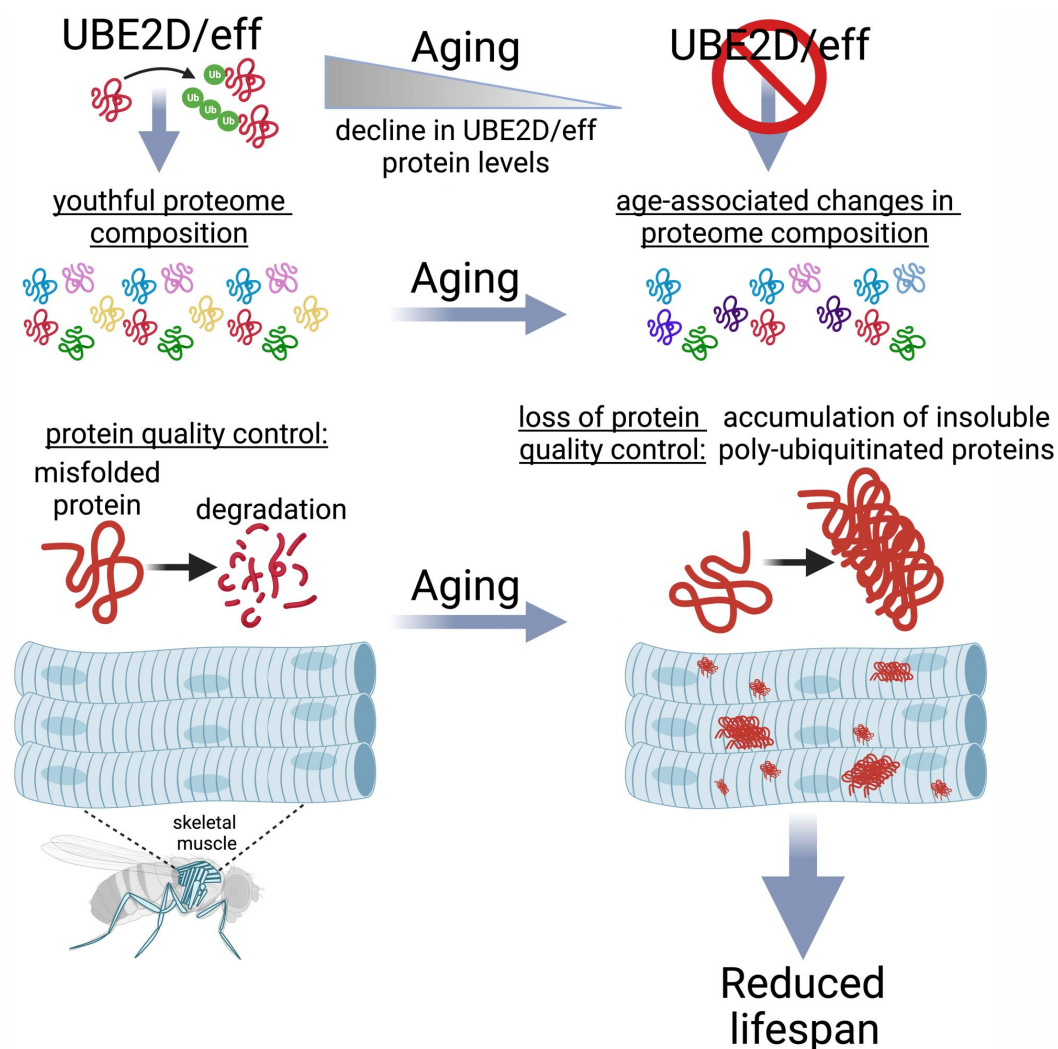


Fig. 7.

The E2 enzyme UBE2D/eff is necessary to preserve proteostasis and to maintain a youthful proteome composition in skeletal muscle during aging

The ubiquitin-conjugating enzyme UBE2D/eff has a key role in proteostasis in skeletal muscle: UBE2D/eff levels decline during aging and reproducing such UBE2D/eff knockdown from a young age causes a loss in protein quality control and a consequent precocious surge in the levels of insoluble poly-ubiquitinated proteins, which normally accumulate only in old age because of age-associated defects in proteostasis. Proteomics surveys indicate that UBE2D/eff knockdown rewires the proteome similar to aging, and that the UBE2D/eff^{RNAi}-induced changes are rescued by transgenic expression of its human homolog UBE2D2. Altogether, these findings indicate that UBE2D/eff is necessary to maintain a youthful proteome and to ensure muscle protein quality control during aging.

knockdown compromises protein quality control also in *Drosophila* skeletal muscle (**Fig. 3** [4](#)), as indicated by the accumulation of insoluble poly-ubiquitinated proteins, and this is in part re-established via overexpression of one of its human homologs, UBE2D2.

Considering that UBE2D/eff is one out of 21 different E2 ubiquitin-conjugating enzymes in *Drosophila*, it is unlikely that UBE2D/eff RNAi alone reduces the total ubiquitination capacity of the cell. In fact, UBE2D/eff knockdown paradoxically increases the levels of poly-ubiquitinated proteins, and such a generalized increase may result from an overall derangement in protein quality control. In agreement with this scenario, the protein categories that are modulated by UBE2D/eff^{RNAi} include regulators of proteostasis such as proteasome components and chaperones (**Fig. 5** [4](#)). Therefore, derangement in the levels of a wide range of proteostasis regulators may lead to a generalized loss of protein quality control and to the accumulation of insoluble poly-ubiquitinated proteins upon UBE2D/eff RNAi.

Previously, UBE2D ubiquitin-conjugating enzymes (UBE2D1/2/3/4) were found to contribute to several cellular processes that impact proteostasis. These include the association of UBE2Ds with the E3 ubiquitin ligase CHIP to promote the degradation of misfolded proteins ([27](#) [4](#)). Likewise, UBE2Ds were found to confer resistance to elevated temperatures in yeast by promoting the selective ubiquitination and degradation of misfolded proteins ([31](#) [4](#), [69](#) [4](#)). Moreover, UBE2Ds can interact with the E3 ligase parkin to promote mitophagy ([29](#) [4](#)) and with the E3 ligase RNF138 to mediate DNA repair ([70](#) [4](#)). In addition to these cellular functions, UBE2D also plays a key role in protein import into peroxisomes ([35](#) [4](#), [71](#) [4](#)-[74](#) [4](#)).

We have found that eff/UBE2D knockdown leads to the accumulation of poly-ubiquitinated proteins and that eff/UBE2D is an E2 ubiquitin-conjugating enzyme that is key for maintaining protein quality control during aging. Because eff/UBE2D knockdown impedes eff/UBE2D-mediated ubiquitination, the poly-ubiquitinated proteins that accumulate upon eff/UBE2D RNAi likely consist of proteins different from eff/UBE2D ubiquitination substrates. How eff/UBE2D knockdown deranges proteostasis and promotes the accumulation of poly-ubiquitinated proteins may occur via multiple mechanisms. One possibility is that proteostasis is deranged by eff/UBE2D knockdown because of the previously-reported roles of eff/UBE2D in mitophagy ([29](#) [4](#)) and in the removal of misfolded proteins ([31](#) [4](#), [69](#) [4](#)) in concert with the E3 ubiquitin ligase CHIP ([27](#) [4](#)). Accumulation of such faulty proteins may clog or burden the proteasome and hence generally impair the degradation of poly-ubiquitinated proteins.

In an alternative model, proteostasis and survival are reduced by eff/UBE2D knockdown because of the accumulation of specific UBE2D protein targets that directly perturb these processes. While many substrates of eff/UBE2D ubiquitination accumulate and may play such role, Arc1 and the related Arc2 protein are of particular interest. Specifically, Arc1 and Arc2 protein levels increase upon UBE2D/eff knockdown and, conversely, they are reduced by hUBE2D2. Interestingly, Arc1 upregulation was found to be cytotoxic and to drive neurodegeneration in a *Drosophila* Alzheimer's disease model ([64](#) [4](#)). This suggests that the accumulation of Arc1/2 in response to eff/UBE2D knockdown may derange proteostasis and reduce survival. Arc1 and Arc2 protein levels increase also in response to aging in skeletal muscle (**Fig. 6** [4](#)). More globally, eff RNAi induces many proteomic changes reminiscent of aging: ~70% of the proteins that are significantly modulated by eff RNAi in young age are also consistently regulated by aging (**Fig. 6** [4](#)). These findings, therefore, indicate that UBE2D/eff maintains a youthful proteome and that the age-dependent decline in UBE2D/eff protein levels (**Fig. 3** [4](#)) may contribute to the proteomic changes that occur with aging.

In addition to reshaping the composition of the muscle proteome and impairing proteostasis, we have found that muscle-specific eff/UBE2D knockdown reduces organismal survival: while muscle-specific eff/UBE2D RNAi reduces lifespan, this is partially rescued by concomitant expression of an RNAi-resistant human UBE2D2 transgene which re-establishes the physiological levels of

eff/UBE2D-regulated targets such as Arc1/2. These findings, therefore, reinforce the notion that skeletal muscle is a key tissue that influences systemic aging, and that preserving muscle protein quality control is necessary for optimal organismal survival (25 [↗](#), 47 [↗](#), 57 [↗](#), 75 [↗](#)-78 [↗](#)).

In summary, our study highlights the important role of the ubiquitin-conjugating enzyme eff/UBE2D in maintaining protein quality control during aging, suggesting that interventions that promote UBE2D function may contrast age-related diseases that arise from the loss of proteostasis.

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Author contributions

L. C. H. did most of the experiments and data analysis, with help from K.N., J.J., and A.S.; K.K. and V.P. did the TMT mass spectrometry and corresponding data analyses; J.P. supervised the mass spectrometry studies; F.D. supervised the project and wrote the manuscript.

Competing interests

The authors declare that they have no competing interests.

Data and materials availability

All data are available within the main text, figures, supplementary tables 1-4, and the source data file. The TMT mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository and are accessible with the dataset identifiers PXD042345 and PXD045713.

Materials and methods

Drosophila husbandry and stocks

Flies were kept (~25 flies/tube) at 25°C, 60% humidity, and a 12h/12h light-dark cycle in tubes containing cornmeal/soy flour/yeast fly food. Survival analysis was done at 25°C and the fly food was changed every 2-3 days (79 [↗](#)). All experiments were done with male flies. Fly stocks were obtained from the Bloomington Drosophila Stock Center (BDSC), the Vienna Drosophila Resource Center (VDRC), and the National Institute of Genetics (NIG-Fly, Japan). The following fly stocks were utilized: *MhcF3-Gal4* ((25 [↗](#), 80 [↗](#)) BL#38464), control *UAS-white^{RNAi}* (BL#33623 and v30033), *UAS-luciferase^{RNAi}* (BL#31603), *UAS-eff^{RNAi}* (BL#35431, #7425R-2, and v26012), *UAS-mcherry* (BL#35787), *UAS-hUBE2D2* (BL#76819), and *UAS-mcherry^{RNAi}* (BL#35785). *GMR-Gal4* and *UAS-Htt-72Q-GFP* flies were previously described (41 [↗](#)). The fly stocks used for the screen in Fig. 1 [↗](#) are reported in Supplementary Table 1.

Whole-mount immunostaining of Drosophila skeletal muscle

The immunostaining of flight skeletal muscle was done as previously described (47 [↗](#), 81 [↗](#), 82 [↗](#)). In brief, thoraces were dissected, fixed for 30 minutes in PBS with 4% paraformaldehyde and 0.1% Triton X-100 at room temperature, washed >3 times in PBS with 0.1% Triton X-100 at room temperature, and immunostained overnight at 4°C with rabbit anti-poly-ubiquitin (FK2; Enzo Life Sciences #BML-PW8810-0100). After washes with PBS with 0.1% Triton X-100, the samples were incubated with secondary antibodies and Alexa635-phalloidin for 2 hours at room temperature, washed, and mounted in an antifade medium. Subsequently, the samples were imaged on a Nikon C2 confocal microscope. Image analysis was done with ImageJ.

Analysis of pathogenic Huntingtin aggregation

Pathogenic huntingtin-polyQ72-GFP protein aggregates (41 [↗](#)) were imaged with an epifluorescence ZEISS SteREO Discovery V12 microscope with consistent exposure time and settings. The acquired grayscale images were then analyzed in an automated manner by using Cell Profiler 3.0.0 (cellprofiler.org [↗](#)) to determine the number and/or total area of protein aggregates (Huntingtin-polyQ72-GFP speckles) normalized by the retinal tissue area. Background fluorescence not corresponding to Huntingtin-polyQ72-GFP aggregates was excluded by thresholding. This analysis was done with male flies after aging at 25°C for 30 days; the same conditions were applied to all the samples and the respective controls in any given experiment. The raw values for each set of flies were normalized by the corresponding raw values of the negative control RNAi intervention from the same fly stock collection.

Western blots for detergent-soluble and -insoluble fractions

Western blots for detergent-soluble and insoluble fractions were done as before (47 [↗](#), 56 [↗](#), 83 [↗](#)). In brief, thoraces were dissected from 20 male flies/sample and homogenized in ice-cold PBS with 1% Triton X-100 containing protease and phosphatase inhibitors. Homogenates were centrifuged at 14,000 rpm at 4°C and supernatants collected (Triton X-100 soluble fraction). The remaining pellet was washed in ice-cold PBS with 1% Triton X-100. The pellet was then resuspended in RIPA buffer containing 8M urea and 5% SDS, centrifuged at 14,000 rpm at 4°C. The supernatants (Triton X-100 insoluble fraction) were collected and analyzed on 4-20% SDS-PAGE with anti-ubiquitin (Cell Signaling Technologies P4D1, #3936) and anti-Ref(2)P/p62 (Abcam #178840) antibodies.

Western blots

For the western blot analysis of flies that express Huntingtin-polyQ72-GFP, 30-day-old flies (n=5/sample) were homogenized in RIPA buffer containing 8M urea and 5% SDS, and centrifuged at 14,000 rpm for 10 minutes at 4°C. The supernatants were collected and analyzed on 4-20% SDS-PAGE with anti-GFP antibodies (Cell Signaling Technologies D5.1, #2956). Anti-α-tubulin antibodies (Cell Signaling Technologies, #2125) were used as loading controls.

qRT-PCR

qRT-PCR was performed as previously described (26, 84–88). Total RNA was extracted with the TRIzol reagent (Life Technologies) from *Drosophila* thoraces, consisting primarily of skeletal muscle, from >20 male flies/replicate, followed by reverse transcription with the iScript cDNA synthesis kit (Bio-Rad). qRT-PCR was performed with SYBR Green and a CFX96 apparatus (Bio-Rad). Three biological replicates were used for each genotype and time point. Alpha-tubulin at 84B (*Tub84B*) was used as a normalization reference. Whole flies at 1 week of age were utilized for the qRT-PCR in Fig. 1D to detect *Htt* and *GFP* levels. The comparative C_T method was used for the relative quantitation of mRNA levels. The following qRT-PCR oligos were used:

eff: 5'-CAATAATGGGCCCGCCGGA-3' and 5'-GCGCGTTGTAAAAGCCACTT-3'

Tub84B: 5'-GTTTGTCAAGCCTCATAGCCG-3' and 5'-GGAAGTGTTCACACGCGAC-3'

GFP: 5'-ACGTAAACGGCCACAAGTTC-3' and 5'-CTTCATGTGGTCGGGGTAGC-3'

Htt (exon 1): 5'-CCTGGATCCCTGGTGAGCAA-3' and 5'-GCTGAACCTGTGGCCGTTTA-3'

Protein sample preparation, protein digestion, and peptide isobaric labeling by tandem mass tags

For each TMT *Drosophila* sample, 50 thoraces (consisting primarily of skeletal muscle) from 1-week-old and 8-weeks-old male w^{1118} flies (Fig. 3) and from 2-weeks-old male flies of the genotypes indicated (Fig. 5) were collected and homogenized in 8M urea lysis buffer (50 mM HEPES, pH 8.5, 8 M urea) (89). After homogenization with zirconium beads in a NextAdvance bullet blender, 0.5% sodium deoxycholate was added to the tissue homogenates, which were then pelleted to remove fly debris. The resulting supernatant was submitted for TMT mass spectrometry and digested with LysC (Wako) at an enzyme-to-substrate ratio of 1:100 (w/w) for 2 hours in the presence of 1 mM DTT. Following this, the samples were diluted to a final 2 M Urea concentration with 50 mM HEPES (pH 8.5), and further digested with trypsin (Promega) at an enzyme-to-substrate ratio of 1:50 (w/w) for at least 3 hours. The peptides were reduced by adding 1 mM DTT for 30 min at room temperature (RT) followed by alkylation with 10 mM iodoacetamide (IAA) for 30 minutes in the dark at RT. The unreacted IAA was quenched with 30 mM DTT for 30 minutes. Finally, the digestion was terminated and acidified by adding trifluoroacetic acid (TFA) to 1%, desalted using C18 cartridges (Harvard Apparatus), and dried by speed vac. The purified peptides were resuspended in 50 mM HEPES (pH 8.5) and labeled with 16-plex Tandem Mass Tag (TMT) reagents (ThermoScientific) following the manufacturer's recommendations and our optimized protocol (55).

Two-dimensional HPLC and mass spectrometry

The TMT-labeled samples were mixed equally, desalted, and fractionated on an offline HPLC (Agilent 1220) by using basic pH reverse-phase liquid chromatography (pH 8.0, XBridge C18 column, 4.6 mm × 25 cm, 3.5 μm particle size, Waters). The fractions were dried and resuspended in 5% formic acid and analyzed by acidic pH reverse phase LC-MS/MS analysis. The peptide samples were loaded on a nanoscale capillary reverse phase C18 column (New objective, 75 μm ID × ~25 cm, 1.9 μm C18 resin from Dr. Maisch GmbH) by an HPLC system (Thermo Ultimate 3000) and eluted by a 60-min gradient. The eluted peptides were ionized by electrospray ionization and detected by an inline Orbitrap Fusion mass spectrometer (ThermoScientific). The mass spectrometer was operated in data-dependent mode with a survey scan in Orbitrap (60,000 resolution,

- $\times 10^6$ AGC target and 50 ms maximal ion time) and MS/MS high-resolution scans (60,000 resolution,
- $\times 10^5$ AGC target, 120 ms maximal ion time, 32 HCD normalized collision energy, 1 m/z isolation window, and 15 s dynamic exclusion).

MS data analysis

The MS/MS raw files were processed by the tag-based hybrid search engine, JUMP (90 [DOI](#)). The raw data were searched against the UniProt human and *Drosophila* databases concatenated with a reversed decoy database for evaluating false discovery rates. Searches were performed by using a 15-ppm mass tolerance for both precursor and product ions, fully tryptic restriction with two maximal missed cleavages, three maximal modification sites, and the assignment of *a*, *b*, and *y* ions. TMT tags on Lys and N-termini (+304.20715 Da) were used for static modifications and Met oxidation (+15.99492 Da) was considered as a dynamic modification. Matched MS/MS spectra were filtered by mass accuracy and matching scores to reduce protein false discovery rate to ~1%. Proteins were quantified by summing reporter ion intensities across all matched PSMs using the JUMP software suite (91 [DOI](#)). Categories enriched in protein sets were identified with DAVID (92 [DOI](#)).

Quantification and statistical analyses

All experiments were performed with biological triplicates unless otherwise indicated. The unpaired two-tailed Student's *t*-test was used to compare the means of two independent groups to each other. One-way ANOVA with Tukey's post hoc test was used for multiple comparisons of more than two groups of normally distributed data. Survival data was analyzed with OASIS 2 (93 [DOI](#)) by using log-rank tests. The "*n*" for each experiment can be found in the figures and/or figure legends and represents independently generated samples, including individual flies for lifespan assays, and batches of flies or fly thoraces for other assays. Bar graphs represent the mean $\pm$ SEM or $\pm$ SD as indicated in the figure legend. Throughout the figures, asterisks indicate a significant *p*-value (**p*<0.05). Statistical analyses were done with Excel and GraphPad Prism.

Description of the Supplementary Data

Supplementary Table 1. RNAi screen for E2 ubiquitin-conjugating enzymes and associated E3 ubiquitin ligases that regulate pathogenic huntingtin-polyQ72-GFP protein aggregates in *Drosophila*. Each E2/E3 was analyzed with multiple RNAi lines, and *n*=5 biological replicates were examined for each RNAi line.

Supplementary Table 2. Mass spectrometry analysis of age-induced protein changes in *Drosophila* skeletal muscle. Tandem mass tag (TMT) mass spectrometry identifies age-related protein changes in the skeletal muscle (thoraces) of old vs. young *w¹¹¹⁸* male flies (8 vs. 1 week old). A tab also reports the list of age-regulated deubiquitinating enzymes (DUBs), E1s, E2s, and E3 enzymes.

Supplementary Table 3. Mass spectrometry analysis of protein changes induced by eff/UBE2D knockdown in *Drosophila* skeletal muscle, and their rescue by co-expression of human UBE2D2. Tandem mass tag (TMT) mass spectrometry identifies protein changes induced in skeletal muscle by eff RNAi (compared to control interventions) and their rescue by co-expression of hUBE2D2 (the human homolog of eff).

Supplementary Table 4. Cross-comparison of proteomic changes induced by aging vs. eff RNAi in skeletal muscle. Cross-comparison of the mass spectrometry data from Supplementary Tables 2 and 3.

Supplementary Table 5. Source data and full scans of western blots.

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Reviewer #1 (Public Review):

In this study, Hunt et al investigated the role of the ubiquitin-conjugating enzyme UBE2D/effete (eff) in maintaining proteostasis during aging. Utilizing *Drosophila* as a model, the researchers observed diverse roles of E2 ubiquitin-conjugating enzymes in handling the aggregation-prone protein huntingtin-polyQ in the retina. While some E2s facilitated aggregate assembly, UBE2D/eff and other E2s were crucial for degradation of htt-polyQ. The study also highlights the significance of UBE2D/eff in skeletal muscle, showing that declining levels of eff during aging correlate with proteostasis disruptions. Knockdown of eff in muscle led to accelerated accumulation of poly-ubiquitinated proteins, shortened lifespan, and mirrored proteomic changes observed in aged muscles. The introduction of human UBE2D2, analogous to eff, partially rescued the deficits in lifespan and proteostasis caused by eff-RNAi expression in muscles.

Comments on revised version:

In this revised manuscript, the authors have addressed some of my concerns, yet several significant caveats remain unaddressed.

One major concern stems from the unexpected outcome observed in the UBE2D/eff loss-of-function experiment. Despite its known role as a ubiquitin-conjugating enzyme (E2), reducing UBE2D/eff levels led to an increase in poly-ubiquitinated proteins and p62 accumulation, suggesting a more complex and multifaceted phenotype seemingly unrelated to the expected role of UBE2D/eff. The authors proposed that an overall disruption of protein quality control, indirectly caused by effRNAi, could explain these phenotypes. However, while the authors noted that effRNAi does not affect proteasome activity, they have not explored other possibilities, leaving a mechanistic explanation still missing.

Furthermore, the comparative analysis of the old versus young proteome identified 10 out of 21 E2 enzymes, suggesting that other E2s may also contribute to age-related changes in proteostasis and lifespan. In this context, the authors mentioned that overexpression of human UBE2D2 in skeletal muscle does not influence lifespan, indicating that the reduced Eff levels observed during aging may not necessarily contribute to the aging phenotype. At this point, I believe the manuscript remains largely descriptive.

<https://doi.org/10.7554/eLife.94739.2.sa2>

Reviewer #2 (Public Review):

The authors screened 21 E2 enzymes for their role in HTTExon1Q72-mCherry (HTT) aggregation in the *Drosophila* eye. They identified UBE2D, whose knockdown leads to increased HTT aggregation that can be rescued by ectopic expression of the human homolog. The protein levels of UBE2D decrease with aging and knockdown of UBE2D leads to an accumulation of ubiquitinated proteins and a shortened lifespan that can be rescued by ectopic expression of the human homolog. Knockdown of UBE2D leads to proteomic changes with up- and down-regulated proteins that include both components of the proteostasis network.

Comments on revised version:

The authors have not addressed a single critical point experimentally. Their explanations are not resolving my concerns and hence the following critical points remain:

- The readout of HTT aggregation (with methods that are not suitable) as proxy for the role of UBE2D in proteostasis is not convincing.
- UBE2D knockdown increases the number of HTT foci (Fig. 1A), but the quantification is less convincing as depicted in Fig. 1B and other E2 enzymes show a stronger effect (e.g. Ubc6 that is only studied in Figs. 1 + 2 without an explanation and Ubc84D). It does not help or add anything to this study that the authors refer to a previous publication. This review assesses this manuscript.
- The quantification of the HTT fluorescence cannot be used as proxy for HTT aggregation. The authors should assess HTT aggregation by e.g. SDD-AGE, FRAP, filter retardation etc. The quantification of the higher MW species of HTT in the SDS-PAGE is not ideal either as this simply reflects material that is stuck in the wells that could not enter the gel. Aggregation and hence high MW size could be one reason, but it can also be HTT trapped in cell debris etc. This point is critical and I disagree with the response of the authors.

- Does UBE2D ubiquitinate HTT? And thus, is HTT accumulation a suitable readout for the functional assessment of the E2 enzyme UBE2D? The authors state that UBE2D does not ubiquitinate HTT. Thus, HTT accumulation is an indirect consequence of perturbed proteostasis. There are certainly better readouts for the role of UBE2D once they have identified substrates.
- The proteomic analyses could help to identify potential substrates for UBE2D. I think it is a missed chance to not follow up on the proteomic analysis to identify substrates and define the role of UBE2D in maintaining proteostasis.
- Are there mutants available for UBE2D or conditional mutants? One caveat of RNAi are: first not complete knockdown and second, variable knockdown efficiencies that increases variability. So mutants are available and yet the authors refuse to use those.
- The analysis of the E3 enzymes does not add anything to this manuscript and the author's response that this manuscript is a follow-up study on a previous publication of the lab is certainly not a valid argument.
- The manuscript remains at this stage rather descriptive.

<https://doi.org/10.7554/eLife.94739.2.sa1>

Reviewer #3 (Public Review):

This is an interesting paper that defines E2 and E3 genes in *Drosophila* that can impact the accumulation of the Q72-GFP protein in the fly eye. The authors then focus on the *eff* gene, showing which human homolog can rescue fly knockdown. They extend to skeletal muscle during natural aging to show that *eff* by TMT mass spec decreases with age normally in the fly muscle and that there is a significant overlap of proteins that are disrupted with *eff* knockdown in young animals in muscle vs aged animals normally in muscle.

Overall these data suggest that *eff* decrease with age may contribute to the increase in ubiquitinated proteins in muscle with age, and that upregulation of *eff* activity might be of interest to extend lifespan. Because *eff* function can be performed by a human homologue the findings may also apply to human situations of aging.

These data are overall interesting and of relevance for those interested in neurodegenerative disease and aging.

<https://doi.org/10.7554/eLife.94739.2.sa0>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

In this study, Hunt et al investigated the role of the ubiquitin-conjugating enzyme UBE2D/effete (eff) in maintaining proteostasis during aging. Utilizing Drosophila as a model, the researchers observed diverse roles of E2 ubiquitinconjugating enzymes in handling the aggregation-prone protein huntingtin-polyQ in the retina. While some E2s facilitated aggregate assembly, UBE2D/eff and other E2s were crucial for degradation of hL-polyQ. The study also highlights the significance of UBE2D/eff in skeletal muscle,

showing that declining levels of eff during aging correlate with proteostasis disruptions. Knockdown of eff in muscle led to accelerated accumulation of poly-ubiquitinated proteins, shortened lifespan, and mirrored proteomic changes observed in aged muscles. The introduction of human UBE2D2, analogous to eff, partially rescued the deficits in lifespan and proteostasis caused by eff-RNAi expression in muscles.

The conclusions of this paper are mostly well supported by data, although a more precise mechanistic explanation of phenotypes associated with UBE2D/eff deficiency would have strengthened the study. Additionally, some aspects of image quantification and data analysis need to be clarified and/or extended.

We thank reviewer #1 for the thoughtful assessment of our work. We have amended the discussion to better explain the phenotypes associated with UBE2D/eff deficiency. We have also improved the methods describing the procedures for image quantification and data analysis.

Reviewer #2 (Public Review):

Important findings:

- Knockdown of UBE2D increases HTT aggregation.
- Knockdown of UBE2D leads to an accumulation of ubiquitinated proteins and reduces the lifespan of *Drosophila*, which is rescued by an ectopic expression of the human homolog.
- UBE2D protein levels decline with aging.
- UBE2D knockdown is associated with an up- and downregulation of several different cellular pathways, including proteostasis components.

Thank you for reviewing our manuscript.

Caveats:

- The readout of HTT aggregation (with methods that are not suitable) as a proxy for the role of UBE2D in proteostasis is not convincing. It would probably improve the manuscript to start with the proteomic analysis of UBE2D to demonstrate that its protein levels decrease with aging. The authors could then induce UBE2D in aged animals to assess the role of UBE2D in the proteome with aging.

While presenting the data in a different order would be possible, we prefer to keep the current order in which from a general screen with a proteostasis readout (HTT aggregates; see the answer below for a discussion on the methods) we proceed to identify a candidate (UBE2D) which is then studied in more detail with additional focused analyses in the retina and skeletal muscle during aging. Concerning the induction of UBE2D in aged animals, our analyses in Figure 4E demonstrate that muscle-specific induction of UBE2D2 throughout life does not increase lifespan alone: this could be explained by UBE2D2 only partially recapitulating the function and substrate diversity of *Drosophila* eff/UBE2D due to divergence from a single *Drosophila* UBE2D enzyme (eff) to multiple UBE2D enzymes in humans (UBE2D1/2/3/4).

- UBE2D knockdown increases the number of HTT foci (Figure 1A), but the quantification is less convincing as depicted in Figure 1B, and other E2 enzymes show a stronger effect (e.g. Ubc6 that is only studied in Figures 1 and 2 without an explanation and Ubc84D).

The graph is hard to interpret. What is the sample size and which genetic conditions show a significant change? P values and statistical analyses are missing.

The full data underlying this genetic screen is reported in Supplementary Table 1. The role of UBC6/UBE2A/B is thoroughly examined in Hunt et al 2021 (PMID: 33658508). We agree that Ubc84D has an important effect and that it should be considered for future studies. We have amended the legend of Figure 1 to indicate that each data point in the graph represents a single RNAi line targeting the corresponding gene. The mean of 5 biological replicates is shown for each RNAi, with each biological replicate representing a single eye imaged from a distinct fly. Therefore, the data points that do not show large magnitude changes may indicate RNAi lines that were not effective at knocking down the target protein (or that did not affect HTT aggregates). The E2s worth pursuing were identified because of multiple RNAi lines scoring consistently: this is the case of UBC6 (studied previously in PMID: 33658508) and eff/UBE2D (pursued in this study). This screen was therefore utilized to identify and select candidate genes (i.e. eff/UBE2D) for more in-depth studies on proteostasis.

- The quantification of the HTT fluorescence cannot be used as a proxy for HTT aggregation. The authors should assess HTT aggregation by e.g. SDD-AGE, FRAP, filter retardation, etc. The quantification of the higher MW species of HTT in the SDS-PAGE is not ideal either as this simply reflects material that is stuck in the wells that could not enter the gel. Aggregation and hence high MW size could be one reason, but it can also be HTT trapped in cell debris, etc.

We agree that the use of multiple methods is a good way to assess the impact of E2 enzymes on HTT protein aggregation. In this regard, we estimated HTT aggregates by fluorescence microscopy and by western blot. Microscopy-based analyses demonstrate both the accumulation of the HTT-GFP pathogenic protein into aggregates (HTT polyQ polypeptides aggregating into one spatial region; Fig. 1 and Fig. 2B) as well as their potential cytotoxicity, resulting in the disruption of the ommatidial ultrastructure and cellular degeneration (Fig. 2A). Similar to native gels and filter retardation, we have utilized SDS-PAGE and western blotting of cellular samples isolated with strong chaotropic and denaturing reagents (8M urea plus detergents and reducing reagents used in the lysis). These experimental conditions maintain the higher-order organization of HTT into high-molecular-weight aggregates that are not broken down into individual polypeptides and that therefore do not readily travel through a gel or filter. Therefore, the biochemical methods we have used are equivalent to those proposed by the reviewer. In addition to combining microscopy-based and biochemical approaches to examine the impact of eff/UBE2D on the HTT aggregates, we have analyzed eff/UBE2D during skeletal muscle aging and found consistent phenotypes as those observed in the HTT model: RNAi for eff/UBE2D leads to the accumulation of detergent-insoluble ubiquitinated proteins that associate with protein aggregates.

- Does UBE2D ubiquitinate HTT? And thus, is HTT accumulation a suitable readout for the functional assessment of the E2 enzyme UBE2D?

We propose that the accumulation of HTT in response to eff/UBE2D RNAi may be due to a generalized loss of protein quality control rather than to a direct decline in the ubiquitination of HTT by eff/UBE2D. In a previous study that examined the UBE2D interactome (Hunt et al. 2023; PMID: 37963875), we did not find an interaction between UBE2D and HTT, suggesting that HTT may not be directly modulated by eff/UBE2D via ubiquitination.

- The proteomic analyses could help to identify potential substrates for UBE2D.

The proteomic analyses in Figure 5 identify several proteins that are modulated by RNAi for eff and by its human homolog, UBE2D2. Such eff/UBE2D2-modulated proteins may indeed be

potential substrates for UBE2D-mediated ubiquitination. For example, this is the case for Pex11 and Pex13, which were found to be upregulated upon UBE2D RNAi also in human cells, where they are ubiquitinated in a UBE2D-dependent manner (Hunt et al. 2023; PMID: 37963875).

- Are there mutants available for UBE2D or conditional mutants? One caveat of RNAi is: first not complete knockdown and second, variable knockdown efficiencies that increase variability.

There are potential hypomorphic alleles of eff/UBE2D that may be available, but they would present the same caveats of incomplete loss of eff/UBE2D function as RNAi. Given the strong phenotype that we find with partial eff knockdown, a caveat of full eff/UBE2D knockout is that this could be lethal.

- The analysis of the E3 enzymes does not add anything to this manuscript.

The analysis of E3 enzymes relates to our recent publication (Hunt et al. 2023; PMID: 37963875) that reports the physical interactions between E2 and E3 enzymes. Analysis of these E2-E3 pairs in the genetic screen in Fig.1 therefore follows this IP-MS study to provide insight into the functional interaction between these E2-E3 pairs in proteostasis.

- Figure 2B: the fluorescence intensities in images 2 and 4 are rather similar, yet the quantification shows significant differences.

Please note that some of the GFP fluorescence in image 4 is not punctate, but rather diffuse fluorescence that is not related to HTT-GFP aggregates. Our image quantitation methods utilized thresholding to identify GFP-positive puncta while eliminating background fluorescence not corresponding to HTT-GFP puncta.

- The proteomic analyses could provide insights into the functional spectrum of UBE2D or even the identification of substrates. Yet apart from a DAVID analysis, none of the hits were followed up. In addition, only a few hits were labelled in the volcano plots (Figure 5). On what basis did the authors select those?

Please see the previous answer above regarding the identification of eff/UBE2D protein substrates from our proteomic analysis in Fig. 5. Only some of the top-regulated hits could be labeled in Fig.5 to avoid overcrowding.

- The manuscript remains at this stage rather descriptive.

Our study has demonstrated a key role for the eff/UBE2D ubiquitin-conjugating enzyme in regulating protein quality control during aging in the *Drosophila* retina and skeletal muscle. Our study has identified key proteins that are modulated by eff/UBE2D RNAi in *Drosophila* muscle, that are rescued by expression of human UBE2D2, and that may underlie the accelerated decline in proteostasis that occurs upon eff/UBE2D RNAi. While more could be known about the regulation of these eff/UBE2D-modulated proteins in *Drosophila*, we have previously demonstrated that some of the proteins that are upregulated by UBE2DRNAi in human cells (e.g. some peroxins) are indeed direct ubiquitination targets of UBE2D via associated E3 ubiquitin ligases (Hunt et al. 2023; PMID: 37963875).

Reviewer #3 (Public Review):

*This is a potentially quite interesting paper that defines E2 and E3 genes in *Drosophila* that can impact the accumulation of the Q72-GFP protein in the fly eye. The authors then*

focus on the eff gene, showing which human homolog can rescue fly knockdown. They extend to skeletal muscle, from the hL protein, to show that eff by TMT mass spec decreases with age normally in the fly muscle and that there is a significant overlap of proteins that are disrupted with eff knockdown in young animals in muscle vs aged animals normally in muscle.

Overall these data suggest eff decrease with age may contribute to the increase in ubiquitinated proteins in muscle with age, and that upregulation of eff activity might be of interest to extending lifespan. Because eff function can be performed by a human homologue, the findings may also apply to human situations of aging.

These data are overall interesting and are of relevance for those interested in neurodegenerative disease and aging, although a number of points from the figures seem confusing and need more explanation or clarity.

Thank you for reviewing our manuscript, we have improved the explanations and clarity of the manuscript.

Recommendations for the authors:

We would like to keep the manuscript title as it is currently to report the partial overlap in the proteomic changes induced by aging and effRNAi (Fig. 6).

Reviewer #1 (Recommendations For The Authors):

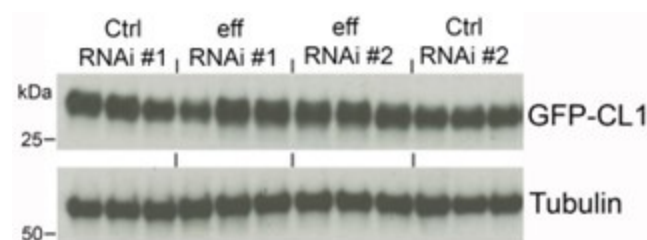
(1) A significant concern arises from the unexpected outcome observed in the UBE2D/eff loss-of-function experiments. Despite its role as a ubiquitin-conjugating enzyme (E2), the reduction in UBE2D/eff levels paradoxically increased polyubiquitinated proteins and p62 accumulation, presenting a more intricate and seemingly unrelated phenotype to its anticipated function.

eff/UBE2D represents one out of 21 different Drosophila E2 ubiquitin-conjugating enzymes and therefore eff RNAi alone is unlikely to reduce the total pool of ubiquitinated proteins. The generalized increase in insoluble polyubiquitinated proteins results from an overall derangement of protein quality control caused by effRNAi. In agreement with this scenario, the protein categories that were found to be modulated by effRNAi (Fig. 5) include proteins associated with protein quality control such as proteasome components and chaperones. Therefore, derangement in the levels of a wide range of regulators of proteostasis may lead to a generalized loss of protein quality control upon effRNAi.

I believe elucidating the mechanisms underlying the impact of UBE2D/eff deficiency on the observed phenotypes would contribute to a more comprehensive understanding of the study's implications. For instance, investigating whether the loss of UBE2D/eff influences muscle proteostasis by impeding proteasome assembly or function, modulating autophagy, etc.

We have previously utilized luciferase assays to measure the proteolytic activity of the proteasome in human cells treated with siRNAs targeting UBE2D1/2/3/4 but found no effect of UBE2D knockdown compared to control nontargeting siRNAs (Hunt et al. 2023; PMID: 37963875). In Drosophila muscles, we have examined the levels of GFP-CL1 (a GFP fused with a proteasomal degron) and found that effRNAi does not impact GFP-CL1 levels (data shown in author response image 1). Overall, these results suggest that effRNAi reduces protein quality control without affecting proteasome activity.

Author response image 1.



(2) Related to Figures 1B-C: It is not clear to this reviewer the quantification methodology used in the experiment. Does each point represent the Average $\pm$ SD for each replicate? If so, it appears that not all cases align with the $n=5$ as indicated in the figure legend. Additionally, how many animals per replicate were quantified?

We have amended the legend of Figure 1 to indicate that each data point in the graph represents a single RNAi line targeting the corresponding gene. The mean of 5 biological replicates is shown for each RNAi line, with each biological replicate representing a single eye imaged from a distinct fly. Therefore, the data points that do not show large magnitude changes may indicate RNAi that were not effective at knocking down the target protein (or with no effect on HTT aggregates).

(3) Related to the previous point: The analysis of pathogenic Huntingtin aggregation in the Materials and Methods section lacks information regarding the number of individuals, replicates, etc.

Please see the response above.

(4) Related to Figure 1 B: In the case of eff/UBE2D, it appears that 3 out of 9 replicates demonstrate a significant increase in HL-polyQ aggregates. Considering the strength of this result, it raises questions about whether it justifies using eff for future analyses.

Please see the response to point (2) above. These results indicate that 3 distinct UAS-RNAi lines targeting eff/UBE2D produced the same effect whereas 6 other effRNAi lines did not, possibly because they are less efficacious in knocking down eff/UBE2D. We have now amended the legend of Fig. 1B to better explain these results.

(5) Related to Figure 1 D-E: Could the authors provide clarification regarding the tissue type and animal age utilized in these experiments?

Whole flies were utilized at 1 week of age.

(6) Related to Figure 3: Incorporating the normal accumulation of poly-ubiquitinated proteins during aging could provide context to better interpret the effect of eff/UBE2D KD at 3 weeks of age.

Several papers from us and others have previously demonstrated a progressive increase in the insoluble levels of poly-ubiquitinated proteins during aging in *Drosophila* skeletal muscle (PMID: 36640359; PMID: 31249065; PMID: 33773104; PMID: 33658508; PMID: 24092876; PMID: 21111239; PMID: 24244197; PMID: 25199830; PMID: 28878259; PMID: 36213625). Our analyses now indicate that such age-related loss of protein quality control is accelerated by eff/UBE2D knockdown.

(7) Related to Figure 3: Would it be possible for the authors to include a list or table detailing the specific E2, deubiquitinating enzymes, and E3s identified in the comparative analysis of the old vs young proteome? This would provide a clear reference for the identified regulatory proteins involved in the age-related proteomic changes.

We have added a tab to Supplementary Table 2 to report the list of age-regulated deubiquitinating enzymes (DUBs) and E1, E2, and E3 enzymes.

(8) Related to Figures 3 and 4: Given that the comparative analysis of the old versus young proteome identified 10 out of 21 E2 ubiquitin-conjugating enzymes, exploring the impact of eff/UBE2D overexpression becomes pivotal to understanding its role in age-related changes in proteostasis and lifespan. Conducting an experiment involving eff overexpression could provide valuable insights into whether restoring eff levels mitigates aging-related phenotypes.

Although we have not done this experiment with eff overexpression, Fig. 4E reports that the overexpression of human UBE2D2 in skeletal muscle does not appear to influence lifespan by itself (green line in Fig. 4E), although it can partially rescue the short lifespan of flies with muscle-specific effRNAi (purple line in Fig. 4E).

(9) Providing a more detailed description of the Supplementary Tables would significantly enhance the reader's comprehension of their content.

A description has been added at the end of the methods.

Reviewer #2 (Recommendations For The Authors):

In addition, to the points listed above:

- The title does not reflect the content of the manuscript and should be changed. There is no evidence that UBE2D maintains a "youthful" (needs to be changed as well) proteome. Rather, its expression declines with aging and its depletion leads to an increase of ubiquitinated proteins. This is true for essentially the entire proteostasis network.

While proteostasis generally declines with aging, it is incompletely understood what specific components of the proteostasis network are dysregulated with aging. Our study now identifies the E2 ubiquitin-conjugating enzyme eff/UBE2D as a key regulator of proteostasis that is transcriptionally downregulated with aging. Comparison of the proteomic changes induced by aging versus those induced by effRNAi in young age indicates a partial overlap (Fig. 6), indicating that eff/UBE2D is, at least in part, necessary to maintain the proteome composition that is found in young age ("youthful"). On this basis, we would like to keep the current title but have amended the manuscript to indicate that such regulation of the proteome composition is only in part dependent on eff/UBE2D.

- Molecular weight markers are missing for the gels/western blot depicted in Fig 1E, 2C, 3E, and 4A.

Thank you for pointing this out, these have been added.

- Fig. 4A, the Ponceau staining for the detergent insoluble samples shows almost no signal for lane 7 and the data should hence not be analyzed.

The western blot membrane in Fig. 4A shows a reliable signal in all lanes (including lane 7) when probed with antibodies for ubiquitin, Ref(2)P, and tubulin. Therefore, there is no reason

for excluding lane 7 from the analysis. Ponceau S staining is provided as an additional loading control but was not used to normalize the data.

Reviewer #3 (Recommendations For The Authors):

There are a number of confusing or not sufficiently explained points in the figures that require clarity.

In Figure 1, panels B and C, one assumes the gray broad line across means no difference from control. For the genes, many have points that are scattered both above and below that control line. What do the dots and range represent for each gene, and why are the data so scattered. How do the authors explain data ranging from no effect, to a negative effect to a positive effect, all for the same gene? Akt1 and Hsp83 are controls but are not quantitated to appreciate how variable the assay is. Can they explain the figure better, and also why the data for any one gene are so variable?

We have amended the legend of Figure 1 to indicate that each data point in the graph represents a single RNAi line targeting the corresponding gene. The mean of 5 biological replicates is shown for each RNAi line, with each biological replicate representing a single eye imaged from a distinct fly. Therefore, the data points that do not show large magnitude changes may indicate RNAi lines that were not effective at knocking down the target protein (or that did not affect HTT aggregates). Therefore, the variability in the analysis of a single gene arises because different RNAi lines targeting that gene may have different efficacy. RNAi lines for Akt1 and Hsp83 are merely used as controls (these have been quantified in Jiao et al. 2023; PMID: 36640359).

In Figure 2A, it is not clear which animals have the hL-Q72-GFP (which eyes are "rough eyes"?). Also, do ubc6-RNAi and eff-RNAi have an impact on the normal eye? That is, can they explain the images and genotypes more clearly.

UBC6 and eff RNAi produce these rough eye phenotypes in the absence of HTT-polyQ and these are rescued by the expression of their human homologs. The panel images indicated in bold here below are those that have "rough eye" phenotypes: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 (a green R has been added to these panels in Fig. 2A).

In Figure 2B, panel 3 looks very different from 1 and 4 and yet is not different from them by quantitation. Can they replace it with a more representative panel or is 3 lower (but not significantly so)?

Please note that some of the GFP fluorescence in image 4 is not punctate, but rather diffuse fluorescence that is not related to HTT-GFP aggregates. Our image quantitation methods utilized thresholding to identify GFP-positive puncta while eliminating background fluorescence not corresponding to HTT-GFP puncta.

In Figures 3E and F, it would be helpful in F to put the detergent soluble bar graphs all on the left so that those data are on the left in both E and F, and then detergent-insoluble in E and F to the right. This would make the figure and quantitation easier to follow.

Done.

The same point as above for Figures 4 A and B.

Done.

In Figure 3A, CG7656 is nearly as reduced with age as eff. One wonders if that gene would give a different or similarly overlapping proteome with age as eff. Was CG7656 not focused on because not conserved?

As indicated in Figure 1B, CG7656 is orthologous to UBE2R1 (also called CDC34) and UBE2R2 in humans. In this screen, however, RNAi targeting CG7656 did not appear to influence HTT aggregates and therefore was not selected for further analyses. However, it may play a role in skeletal muscle proteostasis during aging.

In Figure 6, the R2 value correlating age with eff-RNAi is weak. Although they discuss this in the text, it might also be helpful to include Venn diagrams for gene overlaps and the significance to make the argument more clear that there is a significant correlation in proteins up and down to indicate that eff largely recapitulates the changes of aging. Correlating this with proteins that are restored with UBE2D in muscle in a more clear manner may also be helpful for readers interested in aging.

We have amended the text to indicate that this relatively low correlation ($R^2 \sim 0.2$, but corresponding to a consistent regulation of 70% of proteins by aging and effRNAi) could indicate that eff/UBE2D is only in part responsible for maintaining a youthful composition of the muscle proteome during aging. Other changes that occur with aging likely account for non-correlated alterations in protein levels. We have also added Venn diagrams (Fig. 6E) to further display the overlap in protein regulation by aging vs. effRNAi.

In Figure 7, they might indicate that the accumulated insoluble protein is ubiquitinated. That is left out of the figure, although indicated in the legend.

Done.

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